

More Complicated Logarithmic Equations (Part II) Homework

Solve.

$$\textcircled{1} \log_2 x + \log_2 (x-4) = 5$$

$$\boxed{x=8}$$

$$\textcircled{2} 1 = \log_4 (y-3) + \log_4 y$$

$$\boxed{y=4}$$

$$\textcircled{3} \log_3 x + \log_3 (x+8) = 2$$

$$\boxed{x=1}$$

$$\textcircled{4} \log_6 (x-8) + \log_6 (x-3) = 1$$

$$\boxed{x=9}$$

$$\textcircled{5} 1 = \log_5 (y+2) + \log_5 (y+6)$$

$$\boxed{y=-1}$$

$$\textcircled{6} \log_6 (b+4) + \log_6 (b-5) = 2$$

$$\boxed{b=8}$$

$$\textcircled{7} 3 = \log_2 (x^2 + 2x - 27)$$

$$\boxed{x=5 \text{ or } -7}$$

$$\textcircled{8} \log_6 (b^2 + 11b + 54) = 2$$

$$\boxed{b=-2 \text{ or } -9}$$

$$\textcircled{9} 0 = \log_4 (3x^2 + 11x - 3)$$

$$\boxed{x=\frac{1}{3} \text{ or } -4}$$

$$\textcircled{10} \log_2 (7y) + \log_2 (y - \frac{24}{7}) = 4$$

$$\boxed{y=4}$$

$$\textcircled{11} 1 = \log_3 (4x^2 - 5x + 4)$$

$$\boxed{x=\frac{1}{4} \text{ or } 1}$$

$$\textcircled{12} \log_3 (2y+3) + \log_3 y = 2$$

$$\boxed{y=\frac{3}{2}}$$

$$\textcircled{13} \log_5 b + \log_5 (b+3) = 1$$

$$\boxed{b=\frac{-3+\sqrt{29}}{2}}$$

$$\textcircled{14} 0 = \log_4 (5x) + \log_4 (x-2)$$

$$\boxed{x=\frac{5+\sqrt{30}}{5}}$$

$$\textcircled{15} \log_2 (x+1) + \log_2 x = 3$$

$$\boxed{x=\frac{-1+\sqrt{33}}{2}}$$

$$\textcircled{16} \log_5 y + \log_5 (3y+4) = 1$$

$$\boxed{y=\frac{-2+\sqrt{19}}{3}}$$

$$\textcircled{17} 2 = \log_{(x+2)} (7x+2)$$

$$\boxed{x=1 \text{ or } 2}$$

$$\textcircled{18} \log_{(x-1)} (6x^2 - 16x - 15) = 1$$

$$\boxed{x=\frac{7}{2}}$$

$$\textcircled{19} 2 = \log_{(x+3)} (2x^2 - x + 21)$$

$$\boxed{x=4 \text{ or } 3}$$

More Complicated Logarithmic Equations (Part III)

Exponent Rules Learned in Pre-Algebra

I) $x^a \cdot x^b = x^{a+b}$ Example: $x^2 \cdot x^5 = x^{2+5} = x^7$

II) $\frac{x^a}{x^b} = x^{a-b}$, $x \neq 0$ Example: $\frac{x^9}{x^3} = x^{9-3} = x^6$

New Exponent Rules

III) $(xy)^a = x^a \cdot y^a$ Example: $(2m)^4 = 2^4 \cdot m^4$

IV) $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$, $y \neq 0$ Example: $\left(\frac{n}{3}\right)^7 = \frac{n^7}{3^7}$

V) $(x^a)^b = x^{ab}$ Example: $(x^{10})^3 = x^{10 \cdot 3} = x^{30}$

Strange New Exponent Rules

VI) $x^1 = x$ Example: $3^1 = 3$

VII) $x^0 = 1$, $x \neq 0$ Example: $5^0 = 1$

VIII) $x^{-a} = \frac{1}{x^a}$, $x \neq 0$ Example: $x^{-2} = \frac{1}{x^2}$

IX) $x^{\frac{1}{a}} = \sqrt[a]{x}$ Example: $x^{\frac{1}{3}} = \sqrt[3]{x}$

X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$ or $\sqrt[a]{x^b}$ Example: $x^{\frac{5}{7}} = (\sqrt[7]{x})^5$ or $\sqrt[7]{x^5}$

Logarithm Property V

$$\log_k(AB) = \log_k A + \log_k B$$

Logarithm Property VI

$$\log_k\left(\frac{A}{B}\right) = \log_k A - \log_k B$$

Logarithm Property VII

$$\log_k A^B = B \log_k A$$

Solve. Write the Roman numeral of each logarithm property or exponent rule that you use.

Variables in the Base

① $2 \log_x 3 + \log_x 2 = 3$

L VII $\log_x 3^2 + \log_x 2 = 3$ L VII

$$\log_x 9 + \log_x 2 = 3$$

L V $\log_x (9 \cdot 2) = 3$ L VI

$$\log_x 18 = 3$$

$$x^3 = 18$$

$$\sqrt[3]{x^3} = \sqrt[3]{18}$$

$$\boxed{x = \sqrt[3]{18}}$$

② $2 = \log_y 5 - 4 \log_y 2$

$$2 = \log_y 5 - \log_y 2^4$$

$$2 = \log_y 5 - \log_y 16$$

$$2 = \log_y \left(\frac{5}{16}\right)$$

$$y^2 = \frac{5}{16}$$

$$\sqrt{y^2} = \pm \sqrt{\frac{5}{16}}$$

$$y = \pm \frac{\sqrt{5}}{\sqrt{16}}$$

$$\boxed{y = \frac{\sqrt{5}}{4}} \text{ or } y = -\frac{\sqrt{5}}{4}$$

$$\textcircled{3} 5 = \log_x 8 + 2\log_x 2$$

$$\text{L VII} \quad 5 = \log_x 8 + \log_x 2^2 \quad \text{L VIII}$$

$$5 = \log_x 8 + \log_x 4$$

$$\text{L V} \quad 5 = \log_x (8 \cdot 4) \quad \text{L VII}$$

$$5 = \log_x 32$$

$$x^5 = 32$$

$$\sqrt[5]{x^5} = \sqrt[5]{32} \quad \text{E VI}$$

$$\boxed{x = 2}$$

$$\textcircled{4} 4\log_b 3 - \log_b 9 = 1$$

$$\log_b 3^4 - \log_b 9 = 1$$

$$\log_b 81 - \log_b 9 = 1$$

$$\log_b \left(\frac{81}{9}\right) = 1$$

$$\log_b 9 = 1$$

$$b^1 = 9$$

$$\boxed{b = 9}$$

Equations Involving Fractions

$$\textcircled{5} \log_2 X - \log_2 (X+3) = 4$$

$$\text{L VI} \quad \log_2 \left(\frac{X}{X+3}\right) = 4$$

$$2^4 = \frac{X}{X+3}$$

$$16 = \frac{X}{X+3}$$

$$(X+3)16 = \frac{X}{X+3} (X+3)$$

$$16X + 48 = X$$

$$48 = -15X$$

$$-\frac{48}{15} = X$$

$$-\frac{16}{5} \neq X$$

$$\boxed{\text{No solution}}$$

$$\textcircled{6} 1 = \log_4 (Y+5) - \log_4 (Y+2)$$

$$\text{L VI} \quad 1 = \log_4 \left(\frac{Y+5}{Y+2}\right)$$

$$4^1 = \frac{Y+5}{Y+2}$$

$$\text{E VI} \quad 4 = \frac{Y+5}{Y+2}$$

$$(Y+2)4 = \frac{Y+5}{Y+2} (Y+2)$$

$$4Y + 8 = Y + 5$$

$$3Y = -3$$

$$\boxed{Y = -1}$$

$$\textcircled{1} \log_3 X - \log_3 (X-5) = 2$$

$$\text{LVI} \quad \log_3 \left(\frac{X}{X-5} \right) = 2$$

$$3^2 = \frac{X}{X-5}$$

$$9 = \frac{X}{X-5}$$

$$(X-5)9 = \frac{X}{X-5}(X-5)$$

$$9X - 45 = X$$

$$-45 = -8X$$

$$\boxed{45/8 = X}$$

$$\textcircled{8} 1 = \log_8 (Y+3) - \log_8 (Y-1)$$

$$\text{LVI} \quad 1 = \log_8 \left(\frac{Y+3}{Y-1} \right)$$

$$8^1 = \frac{Y+3}{Y-1}$$

$$\text{EVI} \quad 8 = \frac{Y+3}{Y-1}$$

$$(Y-1)8 = \frac{Y+3}{Y-1}(Y-1)$$

$$8Y - 8 = Y + 3$$

$$7Y = 11$$

$$\boxed{Y = \frac{11}{7}}$$

$$\textcircled{9} \quad \frac{8 + \log_3 X}{\log_3 X + 5} = 4$$

$$(\log_3 X + 5) \frac{8 + \log_3 X}{\log_3 X + 5} = 4(\log_3 X + 5)$$

$$8 + \log_3 X = 4\log_3 X + 20$$

$$8 + \log_3 X - \log_3 X = 4\log_3 X + 20 - \log_3 X$$

$$8 = 3\log_3 X + 20$$

$$-12 = 3\log_3 X$$

$$-4 = \log_3 X$$

$$3^{-4} = X$$

$$\frac{1}{3^4} = X$$

$$\boxed{\frac{1}{81} = X}$$

E VIII

$$\textcircled{10} \quad -3 = \frac{\log_4 Y - 5}{7 \log_4 Y + 1}$$

$$(7 \log_4 Y + 1)(-3) = \frac{\log_4 Y - 5}{7 \log_4 Y + 1} (7 \log_4 Y + 1)$$

$$-21 \log_4 Y - 3 = \log_4 Y - 5$$

$$21 \log_4 Y - 21 \log_4 Y - 3 = \log_4 Y - 5 + 21 \log_4 Y$$

$$-3 = 22 \log_4 Y - 5$$

$$2 = 22 \log_4 Y$$

$$\frac{2}{22} = \log_4 Y$$

$$\frac{1}{11} = \log_4 Y$$

$$4^{\frac{1}{11}} = Y$$

$$\boxed{\sqrt[11]{4} = Y}$$

E IX

$$\textcircled{11} \quad \frac{-9 - \log_5 X}{6 + \log_5 X} = -2$$

$$(6 + \log_5 X) \frac{-9 - \log_5 X}{6 + \log_5 X} = -2 (6 + \log_5 X)$$

$$-9 - \log_5 X = -12 - 2 \log_5 X$$

$$2 \log_5 X - 9 - \log_5 X = -12 - 2 \log_5 X + 2 \log_5 X$$

$$-9 + \log_5 X = -12$$

$$\log_5 X = -3$$

$$5^{-3} = X$$

$$\frac{1}{125} = X$$

$$\boxed{\frac{1}{125} = X}$$

E VIII

★ Note:
You must
use
exponent
rule X.

$$\textcircled{12} \quad \frac{5\log_2 b - 4}{9 - \log_2 b} = 8$$

$$(9 - \log_2 b) \frac{5\log_2 b - 4}{9 - \log_2 b} = 8(9 - \log_2 b)$$

$$5\log_2 b - 4 = 72 - 8\log_2 b$$

$$8\log_2 b + 5\log_2 b - 4 = 72 - 8\log_2 b + 8\log_2 b$$

$$13\log_2 b - 4 = 72$$

$$13\log_2 b = 76$$

$$\log_2 b = \frac{76}{13}$$

$$2^{\frac{76}{13}} = b$$

$$\boxed{(\sqrt[13]{2})^{76} = b}$$

EX

Equations Requiring Logarithm Property IV

Logarithm Property IV

$$b^{\log_b X} = X$$

I) $x^a \cdot x^b = x^{a+b}$ II) $\frac{x^a}{x^b} = x^{a-b}$

$$\textcircled{13} \quad 3^{1 + \log_3 X} = 5$$

EI $3^1 \cdot 3^{\log_3 X} = 5$

EVI $3 \cdot 3^{\log_3 X} = 5$

LIV $3 \cdot X = 5$

$$3X = 5$$

$$\boxed{X = \frac{5}{3}}$$

$$\textcircled{14} \quad 3 = 6^{\log_6 X - 2}$$

EII $3 = \frac{6^{\log_6 X}}{6^2}$

LIV $3 = \frac{X}{36}$

$$\boxed{108 = X}$$

$$\textcircled{15} 2 = 7^{\log_7 X} + 2$$

$$\text{EI} \quad 2 = 7^{\log_7 X} \cdot 7^2$$

$$2 = 7^{\log_7 X} \cdot 49$$

$$\text{LIV} \quad 2 = X \cdot 49$$

$$2 = 49X$$

$$\boxed{\frac{2}{49} = X}$$

$$\textcircled{16} 9^{\log_9 X - 1} = 4$$

$$\text{EII} \quad \frac{9^{\log_9 X}}{9^1} = 4$$

$$\text{EVI} \quad \frac{9^{\log_9 X}}{9} = 4$$

$$\text{LIV} \quad \frac{X}{9} = 4$$

$$\boxed{X = 36}$$

Logarithm Property VIII

If $\log_k A = \log_k B$, then $A = B$.

Equations Requiring Logarithm Property VIII

$$\textcircled{17} \log_4 (X+6) = \log_4 (2X)$$

L VIII

$$X+6 = 2X$$

$$\boxed{6 = X}$$

$$\textcircled{18} 2 \log_5 b = \log_5 9$$

L VII

$$\log_5 b^2 = \log_5 9$$

L VIII

$$b^2 = 9$$

$$\sqrt{b^2} = \pm \sqrt{9}$$

$$b = \pm 3$$

$$\boxed{b = 3} \text{ or } b \neq -3$$

$$\textcircled{19} \log_8 (3X) = \log_8 (4-9X)$$

L VIII

$$3X = 4 - 9X$$

$$12X = 4$$

$$\boxed{X = \frac{1}{3}}$$

L VII

$$\textcircled{20} \log_2 6 = 3 \log_2 Y$$

L VIII

$$\log_2 6 = \log_2 Y^3$$

$$6 = Y^3$$

$$\sqrt[3]{6} = \sqrt[3]{Y^3}$$

$$\boxed{\sqrt[3]{6} = Y}$$

$$\textcircled{21} \quad 2\log_3(5x) + \log_3 2 = 3\log_3 4$$

$$\text{L VII} \quad \log_3(5x)^2 + \log_3 2 = \log_3 4^3$$

$$\text{E III} \quad \log_3(5^2 \cdot x^2) + \log_3 2 = \log_3 64$$

$$\log_3(25x^2) + \log_3 2 = \log_3 64$$

$$\text{L V} \quad \log_3[(25x^2)(2)] = \log_3 64$$

$$\log_3[50x^2] = \log_3 64$$

$$\text{L VIII} \quad 50x^2 = 64$$

$$x^2 = \frac{64}{50}$$

$$\sqrt{x^2} = \pm \sqrt{\frac{64}{50}}$$

$$x = \pm \frac{\sqrt{64}}{\sqrt{50}}$$

$$x = \pm \frac{8}{5\sqrt{2}}$$

$$x = \pm \frac{8\sqrt{2}}{5 \cdot 2}$$

$$x = \pm \frac{8\sqrt{2}}{10}$$

$$x = \pm \frac{4\sqrt{2}}{5}$$

$$x = \frac{4\sqrt{2}}{5} \text{ or } x \neq -\frac{4\sqrt{2}}{5}$$

$$\star \textcircled{22} \quad \log_2 35 = \log_2 5 + 3\log_2(3x+1)$$

$$\text{L VII} \quad \log_2 35 = \log_2 5 + \log_2 (3x+1)^3$$

$$\text{L V} \quad \log_2 35 = \log_2 [5(3x+1)^3]$$

$$\text{L VIII} \quad 35 = 5(3x+1)^3$$

$$\frac{35}{5} = \frac{5(3x+1)^3}{5}$$

$$7 = (3x+1)^3$$

$$\sqrt[3]{7} = \sqrt[3]{(3x+1)^3}$$

$$\sqrt[3]{7} = 3x+1$$

$$\sqrt[3]{7} - 1 = 3x$$

$$\boxed{\frac{\sqrt[3]{7}-1}{3} = x}$$

$$\textcircled{23} \log_6 64 = \log_6 2 + 3 \log_6 (2x)$$

$$\text{L VII} \quad \log_6 64 = \log_6 2 + \log_6 (2x)^3$$

$$\text{E III} \quad \log_6 64 = \log_6 2 + \log_6 (2^3 \cdot x^3)$$

$$\log_6 64 = \log_6 2 + \log_6 (8x^3)$$

$$\text{L V} \quad \log_6 64 = \log_6 [(2)(8x^3)]$$

$$\log_6 64 = \log_6 [16x^3]$$

$$\text{L VIII} \quad 64 = 16x^3$$

$$4 = x^3$$

$$\sqrt[3]{4} = \sqrt[3]{x^3}$$

$$\boxed{\sqrt[3]{4} = x}$$

$$\star \textcircled{24} \quad 2 \log_5 (x-2) + \log_5 3 = \log_5 9$$

$$\text{L VII} \quad \log_5 (x-2)^2 + \log_5 3 = \log_5 9$$

$$\text{L V} \quad \log_5 [(x-2)^2(3)] = \log_5 9$$

$$\text{L VIII} \quad 3(x-2)^2 = 9$$

$$\frac{3(x-2)^2}{3} = \frac{9}{3}$$

$$(x-2)^2 = 3$$

$$\sqrt{(x-2)^2} = \pm \sqrt{3}$$

$$x-2 = \pm \sqrt{3}$$

$$x = \pm \sqrt{3} + 2$$

$$\boxed{x = \sqrt{3} + 2} \text{ or } x \neq -\sqrt{3} + 2$$

Logarithms within Logarithms

$$\textcircled{25} \log_4 [15 + \log_3 (x-7)] = 2$$

$$4^2 = 15 + \log_3 (x-7)$$

$$16 = 15 + \log_3 (x-7) \quad \text{EVI}$$

$$1 = \log_3 (x-7)$$

$$3^1 = x-7$$

$$\text{EVI} \quad 3 = x-7$$

$$\boxed{10 = x}$$

$$\textcircled{26} 0 = \log_3 [\log_2 (5x) - 4]$$

$$3^0 = \log_2 (5x) - 4$$

$$1 = \log_2 (5x) - 4$$

$$5 = \log_2 (5x)$$

$$2^5 = 5x$$

$$32 = 5x$$

$$\boxed{32/5 = x}$$

$$\textcircled{27} 1 = \log_2 [\log_7 (2x-3) + 1]$$

$$2^1 = \log_7 (2x-3) + 1$$

$$\text{EVI} \quad 2 = \log_7 (2x-3) + 1$$

$$1 = \log_7 (2x-3)$$

$$7^1 = 2x-3$$

$$\text{EVI} \quad 7 = 2x-3$$

$$\boxed{5 = x}$$

$$\textcircled{28} 3 = \log_4 [61 - \log_2 (7x)]$$

$$4^3 = 61 - \log_2 (7x)$$

$$64 = 61 - \log_2 (7x)$$

$$3 = -\log_2 (7x)$$

$$-3 = \log_2 (7x)$$

$$2^{-3} = 7x$$

$$\frac{1}{2^3} = 7x$$

$$\frac{1}{8} = 7x$$

$$\boxed{\frac{1}{56} = x}$$